

and $C^2(\theta | \mathcal{H}_1) \leq C^2(\theta | \mathcal{H}_0)$ and, therefore, $E(\theta, n) < 0$. This intuition can be made rigorous using the explicit formulas for $E(\theta, n)$ provided by Eq. (32) and Eq. (33) from Appendix A, as is shown in the following example.

4 Example

We consider a Bayesian t -test and for the peri-null Bayes factor use the priors

$$\pi(\delta, \sigma | \mathcal{H}_1) \propto \text{Cauchy}(\delta; 0, \kappa_1)\sigma^{-1} \text{ and } \pi(\delta, \sigma | \mathcal{H}_0) \propto \mathcal{N}(\delta; 0, \kappa_0^2)\sigma^{-1}. \quad (15)$$

Note that $\pi(\delta, \sigma | \mathcal{H}_1)$ is chosen as in the default Bayesian t -test (Jeffreys, 1948; Ly et al., 2016b,a; Rouder et al., 2009), where $\kappa_1 > 0$ denotes the scale parameter of the Cauchy distribution on standardized effect size $\delta = \mu/\sigma$, and $\sigma \propto \sigma^{-1}$ implies that the standard deviation common in both models is proportional to σ^{-1} (for advantages of this choice see Hendriksen et al., 2021; Grünwald et al., 2019). For data-governing parameters $\theta = (\mu, \sigma)$, where μ is the population mean, Theorem 1 shows that as $n \rightarrow \infty$

$$\log \text{BF}_{10}(Y^n; \kappa_0, \kappa_1) \xrightarrow{\mathbb{P}_\theta} \log \left(\frac{\sqrt{2}\kappa_0 \exp(\frac{\mu^2}{2\kappa_0^2\sigma^2})}{\sqrt{\pi}\kappa_1(1 + [\frac{\mu}{\kappa_1\sigma}]^2)} \right) =: v(\theta). \quad (16)$$

Direct calculations show that $\dot{v}(\theta) = 0$ only when $\mu = 0$. Hence, under the alternative $\mu \neq 0$, the logarithm of these peri-null Bayes factor t -tests are asymptotically normal with an approximate variance of

$$\frac{(\mu^4 + 2\mu^2\sigma^2)(\mu^2 + (\kappa_1^2 - 2\kappa_0^2)\sigma^2)^2}{2\kappa_0^4\sigma^4(\mu^2 + \kappa_1^2\sigma^2)^2n}. \quad (17)$$

To characterize the asymptotic mean we also require the bias term $E(\theta, n)$. An application of Eq. (32) and Eq. (33) from Appendix A shows that for the problem at hand the bias term comprises of

$$C^1(\theta | \mathcal{H}_1) = \frac{13\mu^4 + (18 + 2\kappa_1^2)\sigma^2\mu^2 + (\kappa_1^2 - 6)\kappa_1^2\sigma^4}{6(\mu^2 + \kappa_1^2\sigma^2)^2}, \quad (18)$$

$$C^2(\theta | \mathcal{H}_1) = \frac{780\mu^6 + (1110 + 3127\kappa_1^2)\sigma^2\mu^4 + (6020 + 4462\kappa_1^2)\kappa_1^2\sigma^4\mu^2 + (5091\kappa_1^2 - 1426)\kappa_1^4\sigma^6}{-96(\mu^2 + \kappa_1^2\sigma^2)^3}, \quad (19)$$

$$C^1(\theta | \mathcal{H}_0) = \frac{3\mu^4 + 6\sigma^2\mu^2 + \kappa_0^2\sigma^4(2\kappa_0^2 - 6)}{12\kappa_0^4\sigma^4}, \quad (20)$$

$$C^2(\theta | \mathcal{H}_0) = \frac{124\mu^6 + (264 - 2369\kappa_0^2)\sigma^2\mu^4 + (10811\kappa_0^2 - 2218)\kappa_0^2\sigma^4\mu^2 + 2(713 - 5091\kappa_0^2)\kappa_0^4\sigma^6}{192\kappa_0^6\sigma^6}. \quad (21)$$

More concretely, under $\mu = 0.167$ and $\sigma = 1$, $\log \text{BF}_{10}(Y^n; 0.05, 1)$ converges in probability to $\log(10)$. This limit is depicted as the pink dashed horizontal line in the top left subplot of Fig. 1.

This subplot also shows the mean (solid green curve) and the 97.5% and 2.5% quantiles (dotted green curves above and below the solid curve

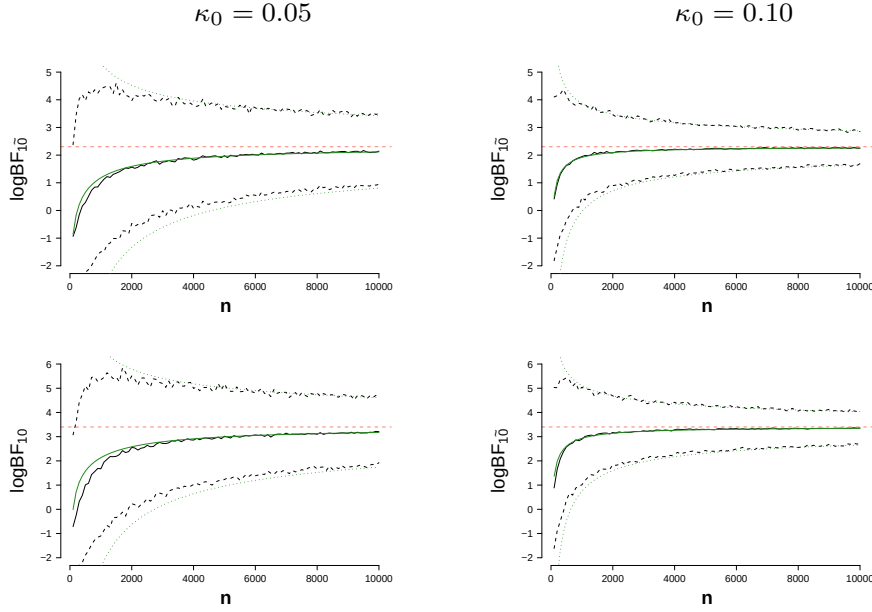


Fig. 1: Under the alternative, the logarithm of the peri-null Bayes factor t -test is asymptotically normal with a mean (i.e., the solid curves) that increases to the limit, e.g., $\log \text{BF}_{10} = \log(10)$ and $\log \text{BF}_{10} = \log(30)$ in the top and bottom row respectively. The black and green curves correspond to the simulated and asymptotic normal sampling distribution respectively. The dotted curves show the 97.5% and 2.5% quantiles of the respective sampling distribution. Note that the convergence to the upper bound is slower when the peri-null is more concentrated, e.g., compare the left to the right column.

respectively) based on the asymptotic normal result of Theorem 2. The black curves represent the analogous quantities based on simulated normal data with $\mu = 0.167$, $\sigma = 1$ based on 1,000 replications at sample sizes $n = 100, 200, 300, \dots, 10000$.

Observe that for small sample sizes, the simulated peri-null Bayes factors are more concentrated on small values. In this regime the concentration of the peri-null prior dominates, and the Laplace approximation of $p(Y^n | \mathcal{H}_{\tilde{0}})$ is still inaccurate.

As expected, the Laplace approximation becomes accurate sooner, whenever the peri-null prior is less concentrated. The top right subplot depicts results of $\log \text{BF}_{10}(Y^n; 0.10, 1)$ under $\mu = 0.314$ and $\sigma = 1$, which converges in probability to $\log(10)$.

Similarly, the asymptotic normal distribution becomes adequate at a smaller sample size for larger population means μ . The bottom left subplot corresponds to $\log \text{BF}_{10}(Y^n; 0.05, 1)$ under $\mu = 0.182$ and $\sigma = 1$, whereas the bottom right

subplot corresponds to $\log \text{BF}_{1\tilde{0}}(Y^n; 0.10, 1)$ under $\mu = 0.348$ and $\sigma = 1$. The logarithms of both peri-null Bayes factors converge in probability to $\log(30)$.

In sum, the plots show that under the alternative hypothesis the asymptotic normal distribution approximates the sampling distribution of the logarithm of the peri-null Bayes factor quite well, and it approximates better when the peri-null prior is less concentrated.

Under the null hypothesis $\mu = 0$, the gradient $\dot{v}(0, \sigma) = 0$, and so is the Hessian, except for the the first entry of $\ddot{v}$, that is,

$$\left. \frac{\partial^2}{\partial \mu^2} v(\mu, \sigma) \right|_{\mu=0} = \frac{\kappa_1^2 - 2\kappa_0^2}{\kappa_0^2 \kappa_1^2 \sigma^2}. \quad (22)$$

As such, $\log \text{BF}_{1\tilde{0}}(Y^n)$ has a shifted asymptotically $\chi^2(1)$ -distribution, i.e.,

$$n \left(\log \text{BF}_{1\tilde{0}}(Y^n; \kappa_0, \kappa_1) - \log \left(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)} \right) - E(\theta, n) \right) \overset{\mathbb{P}_{0,1}}{\rightsquigarrow} \frac{\kappa_1^2 - 2\kappa_0^2}{2\kappa_0^2 \kappa_1^2} Z^2, \quad (23)$$

where $Z \sim \mathcal{N}(0, 1)$.

More concretely, under $\mu = 0$ and $\sigma = 1$, $\log \text{BF}_{1\tilde{0}}(Y^n; 0.05, 1) \xrightarrow{\mathbb{P}_{0,1}} -3.22$, whereas $\log \text{BF}_{1\tilde{0}}(Y^n; 0.10, 1)$ converges in probability to -2.53 . Both cases yield evidence for the null hypothesis, but the evidence is stronger for the peri-null that is more tightly concentrated around 0. The approximation based on the asymptotic $\chi^2(1)$ -distribution (in green) and the simulations (in black) are shown in Fig. 2. In the left subplot, the curves based on the asymptotic $\chi^2(1)$ -

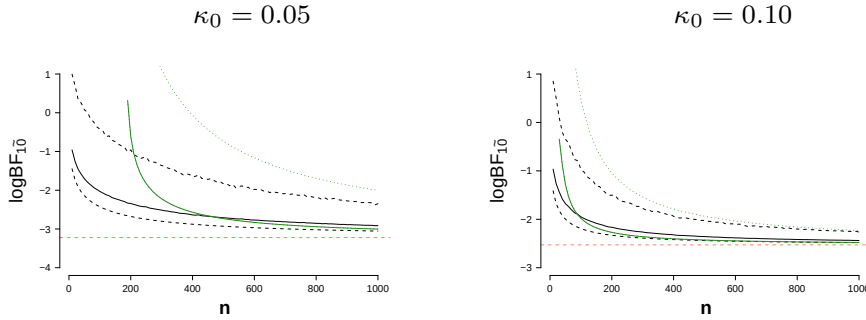


Fig. 2: Under the null, the logarithm of the peri-null Bayes factor t -test has a shifted asymptotically $\chi^2(1)$ -distribution with a mean (i.e., the solid curves) that decreases to the limit, e.g., $\log \text{BF}_{1\tilde{0}} = -3.22$ and $\log \text{BF}_{1\tilde{0}} = -2.53$ in the left and right plot respectively. The black and green curves correspond to the simulated and asymptotic $\chi^2(1)$ sampling distribution respectively. The dotted curves show the 97.5% and 2.5% quantiles of the respective sampling distribution. Note that the convergence to the lower bound is slower when the peri-null is more concentrated, e.g., compare the left to the right plot.

distribution only start from $n = 185$, because only for $n \geq 185$ does $\log(1 +$